CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

LUIS C NUNES & SONS DAIRY #1

29191 HUSMAN RD

GUSTINE CA 95322

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Luis C Nunes & Sons Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
29191 Husman RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X060-X004-XXXX X070-X060-X005-XXXX

B. OPERATORS

Nunes, Louie

Operator name: Nunes, Louie

Telephone no.: Landline: ; Cellular: (209) 652-7647

Mailing Address Number and Street	City	State	Zip Code
735 Ashley CT	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Nunes, Luis C

Legal owner name: Nunes, Luis C

Telephone no.: Landline: (209) 854-3293; Cellular:

Mailing Address Number and Street	City	State	Zip Code
29191 Husman RD	Gustine	CA	95322

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	545	490	115	220
Number under roof	590	205	0	0	0	0
Maximum number	590	205	545	490	115	220
Average number	590	205	545	490	115	220
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 79 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 30,595.60 *tons per reporting period*

Total nitrogen from manure: 358,774.74 *lbs per reporting period*

After ammonia losses (30% loss applied): 251,142.32 *lbs per reporting period*

Total phosphorus from manure: 57,768.18 *lbs per reporting period*

Total potassium from manure: 116,479.11 *lbs per reporting period*

Total salt from manure: 324,941.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 15,772,500 *gallons*

Total nitrogen generated: 94,731.72 *lbs*

Total phosphorus generated: 13,323.78 *lbs*

Total potassium generated: 152,961.47 *lbs*

Total salt generated: 709,239.05 *lbs*

	15,772,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	15,772,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	Separator solids	1,575.00 *ton*	Dry-weight	49.6		21,300.00	5,500.00	19,700.00		56.20
10/18/2023	Separator solids	1,300.00 *ton*	Dry-weight	73.0		29,700.00	9,900.00	42,000.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	54,665.28	15,681.60	60,759.72	892,231.20
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	54,665.28	15,681.60	60,759.72	892,231.20

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
A Silva	61	61	2	both	X070-X060-X034-XXXX X070-X060-X035-XXXX
G Silva	39	39	1	manure	X070-X060-X013-XXXX
Home	87	87	2	process wastewater	X070-X030-X013-XXXX X070-X060-X004-XXXX X070-X060-X005-XXXX
M Silva	77	77	2	manure	X070-X060-X029-XXXX
Stephens	23	23	2	manure	X070-X070-X039-XXXX X070-X070-X040-XXXX
Totals for areas that were used for application	287	287	9		
Totals for areas that were not used for application					
Land application area totals	287	287	9		

B. CROPS AND HARVESTS

A Silva

Field name: A Silva

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,098.00 *ton*	Dry-weight		68.6	19,800.00	3,600.00	23,000.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	223.82	40.69	259.99	915.62

A Silva

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	1,952.00 *ton*	Dry-weight		62.1	14,400.00	2,500.00	11,900.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	349.29	60.64	288.65	1,382.59

G Silva

Field name: G Silva

11/04/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 39 Plant date: 11/04/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	312.00 *ton*	Dry-weight		10.3	32,500.00	2,900.00	23,100.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.00	466.44	41.62	331.53	1,363.44

Home

Field name: Home

Home

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 87 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/11/2023	1,566.00 *ton*	Dry-weight		78.5	16,300.00	3,800.00	26,600.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	126.16	29.41	205.88	890.10

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 87 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	2,784.00 *ton*	Dry-weight		64.9	15,000.00	2,650.00	15,200.00		6.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	336.96	59.53	341.45	1,426.46

M Silva

Field name: M Silva

11/04/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 77 Plant date: 11/04/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/28/2023	200.00 *ton*	Dry-weight		10.4	34,400.00	3,000.00	24,400.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	2.60	160.12	13.96	113.57	414.25

M Silva

05/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 77 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	539.00 *ton*	Dry-weight		56.7	14,500.00	2,700.00	27,500.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.00	87.90	16.37	166.71	763.81

Stephens

Field name: Stephens

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	414.00 *ton*	Dry-weight		73.6	18,050.00	3,700.00	24,800.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	171.55	35.16	235.70	931.39

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	736.00 *ton*	Dry-weight		67.9	12,100.00	2,600.00	13,100.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	248.58	53.41	269.13	1,212.10

NUTRIENT BUDGET

A. LAND APPLICATIONS

A Silva - 10/16/2022: Oats, silage-soft dough

Field name: A Silva

Crop: Oats, silage-soft dough — Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	100.96	18.67	135.25	0.00	280.00 *ton*
Application event totals		100.96	18.67	135.25	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.56	0.00	0.00	159.27	5,821,291.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.94	12.53	152.72	758.44	1,260,000.00 *gal*
Canal	Surface water	0.47	0.00	0.00	132.90	4,857,417.00 *gal*
Application event totals		97.41	12.53	152.72	891.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.41	19.30	98.81	424.77	675,000.00 *gal*
Canal	Surface water	0.47	0.00	0.00	133.60	4,882,782.00 *gal*
Application event totals		105.88	19.30	98.81	558.37	

A Silva - 05/25/2023: Corn, silage

Field name: A Silva

Crop: Corn, silage — Plant date: 05/25/2023

A Silva - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.36	13.17	144.05	655.01	900,000.00 *gal*
Canal	Surface water	1.34	0.00	0.00	383.35	14,011,049.00 *gal*
Application event totals		92.70	13.17	144.05	1,038.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.44	12.74	176.99	821.23	1,035,000.00 *gal*
Canal	Surface water	1.01	0.00	0.00	289.66	10,586,760.00 *gal*
Application event totals		98.46	12.74	176.99	1,110.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.73	11.08	153.90	714.11	900,000.00 *gal*
Canal	Surface water	1.00	0.00	0.00	285.58	10,437,740.00 *gal*
Application event totals		85.73	11.08	153.90	999.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	307.97	11,255,765.00 *gal*
Application event totals		1.08	0.00	0.00	307.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.68	13.30	184.68	856.94	1,080,000.00 *gal*
Canal	Surface water	1.27	0.00	0.00	364.01	13,303,997.00 *gal*
Application event totals		102.95	13.30	184.68	1,220.94	

A Silva - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.21	12.19	169.29	785.52	990,000.00 *gal*
Canal	Surface water	1.29	0.00	0.00	368.52	13,468,870.00 *gal*
Application event totals		94.50	12.19	169.29	1,154.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	340.15	12,432,071.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

G Silva - 11/04/2020: Alfalfa, hay

Field name: G Silva

Crop: Alfalfa, hay

Plant date: 11/04/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.76	0.00	0.00	217.76	5,088,363.00 *gal*
Application event totals		0.76	0.00	0.00	217.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	373.59	8,729,738.00 *gal*
Application event totals		1.31	0.00	0.00	373.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	375.04	8,763,561.00 *gal*
Application event totals		1.31	0.00	0.00	375.04	

G Silva - 11/04/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.13	0.00	0.00	322.01	7,524,593.00 *gal*
Application event totals		1.13	0.00	0.00	322.01	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	313.33	7,321,658.00 *gal*
Application event totals		1.10	0.00	0.00	313.33	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.29	0.00	0.00	369.32	8,630,051.00 *gal*
Application event totals		1.29	0.00	0.00	369.32	

09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	286.44	6,693,274.00 *gal*
Application event totals		1.00	0.00	0.00	286.44	

10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.89	0.00	0.00	254.14	5,938,501.00 *gal*
Application event totals		0.89	0.00	0.00	254.14	

10/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	316.65	105.55	447.78	0.00	770.00 *ton*
Application event totals		316.65	105.55	447.78	0.00	

Home - 10/16/2022: Oats, silage-soft dough

Home - 10/16/2022: Oats, silage-soft dough

Field name: Home

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.40	9.10	110.91	550.77	1,305,000.00 *gal*
Canal	Surface water	0.56	0.00	0.00	159.27	8,302,497.00 *gal*
Application event totals		70.96	9.10	110.91	710.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.47	0.00	0.00	132.90	6,927,792.00 *gal*
Application event totals		0.47	0.00	0.00	132.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.08	17.59	90.06	387.18	877,500.00 *gal*
Canal	Surface water	0.47	0.00	0.00	133.60	6,963,968.00 *gal*
Application event totals		96.55	17.59	90.06	520.78	

Home - 05/25/2023: Corn, silage

Field name: Home

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.68	12.93	141.40	642.97	1,260,000.00 *gal*
Canal	Surface water	1.34	0.00	0.00	383.35	19,982,972.00 *gal*
Application event totals		91.02	12.93	141.40	1,026.32	

Home - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.06	12.43	172.66	801.12	1,440,000.00 *gal*
Canal	Surface water	1.01	0.00	0.00	289.66	15,099,150.00 *gal*
Application event totals		96.07	12.43	172.66	1,090.78	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	285.58	14,886,613.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.63	11.46	159.17	738.53	1,327,500.00 *gal*
Canal	Surface water	1.08	0.00	0.00	307.97	16,053,304.00 *gal*
Application event totals		88.71	11.46	159.17	1,046.50	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.60	11.85	164.56	763.57	1,372,500.00 *gal*
Canal	Surface water	1.27	0.00	0.00	364.01	18,974,553.00 *gal*
Application event totals		91.88	11.85	164.56	1,127.57	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.12	11.65	161.86	751.05	1,350,000.00 *gal*
Canal	Surface water	1.29	0.00	0.00	368.52	19,209,700.00 *gal*
Application event totals		90.41	11.65	161.86	1,119.57	

Home - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	340.15	17,730,987.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

M Silva - 11/04/2020: Alfalfa, hay

Field name: M Silva

Crop: Alfalfa, hay

Plant date: 11/04/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.76	0.00	0.00	217.76	10,046,255.00 *gal*
Application event totals		0.76	0.00	0.00	217.76	

M Silva - 05/25/2023: Sudangrass, silage

Field name: M Silva

Crop: Sudangrass, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.12	0.00	0.00	320.22	14,773,403.00 *gal*
Application event totals		1.12	0.00	0.00	320.22	

M Silva - 05/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	107.35	27.72	99.29	2,832.48	385.00 *ton*
Application event totals		107.35	27.72	99.29	2,832.48	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.31	0.00	0.00	375.04	17,302,414.00 *gal*
Application event totals		1.31	0.00	0.00	375.04	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.13	0.00	0.00	322.01	14,856,247.00 *gal*
Application event totals		1.13	0.00	0.00	322.01	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.10	0.00	0.00	313.33	14,455,582.00 *gal*
Application event totals		1.10	0.00	0.00	313.33	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.29	0.00	0.00	369.32	17,038,819.00 *gal*
Application event totals		1.29	0.00	0.00	369.32	

09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	286.44	13,214,925.00 *gal*
Application event totals		1.00	0.00	0.00	286.44	

M Silva - 05/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.89	0.00	0.00	254.14	11,724,732.00 *gal*
Application event totals		0.89	0.00	0.00	254.14	

Stephens - 10/16/2022: Oats, silage-soft dough

Field name: Stephens

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	219.95	40.67	294.65	0.00	230.00 *ton*
Application event totals		219.95	40.67	294.65	0.00	

10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.56	0.00	0.00	159.27	2,194,913.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

11/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.47	0.00	0.00	132.90	1,831,485.00 *gal*
Application event totals		0.47	0.00	0.00	132.90	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.47	0.00	0.00	133.60	1,841,049.00 *gal*
Application event totals		0.47	0.00	0.00	133.60	

Stephens - 05/25/2023: Corn, silage

Field name: Stephens

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	326.72	84.37	302.18	8,620.59	350.00 *ton*
Application event totals		326.72	84.37	302.18	8,620.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.34	0.00	0.00	383.35	5,282,855.00 *gal*
Application event totals		1.34	0.00	0.00	383.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.01	0.00	0.00	289.66	3,991,729.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.00	0.00	0.00	285.58	3,935,541.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	307.97	4,243,977.00 *gal*
Application event totals		1.08	0.00	0.00	307.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	364.01	5,016,261.00 *gal*
Application event totals		1.27	0.00	0.00	364.01	

Stephens - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.29	0.00	0.00	368.52	5,078,426.00 *gal*
Application event totals		1.29	0.00	0.00	368.52	

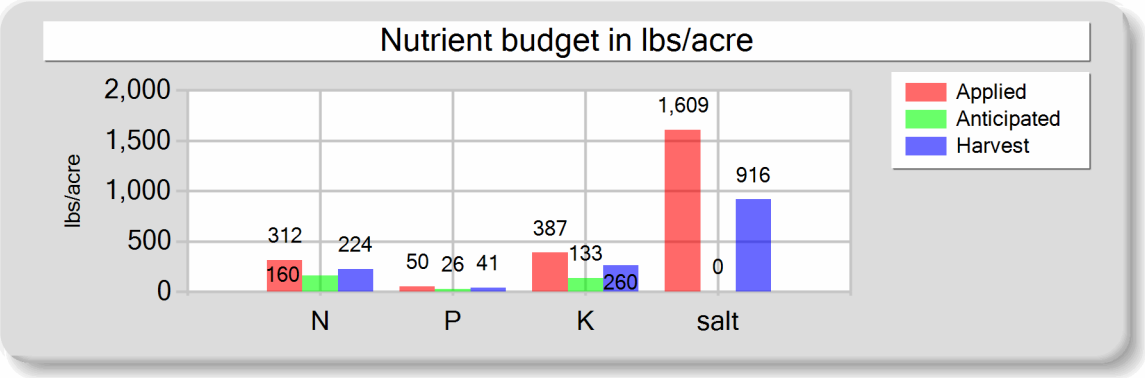
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.19	0.00	0.00	340.15	4,687,502.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

B. NUTRIENT BUDGET

A Silva - 10/16/2022: Oats, silage-soft dough

Field name: A Silva | Crop: Oats, silage-soft dough | Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	100.96	18.67	135.25	0.00
Process wastewater	202.35	31.83	251.53	1,183.21
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.80	50.50	386.78	1,608.98
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	223.82	40.69	259.99	915.62
Nutrient balance	87.98	9.80	126.79	693.36
Applied to removed ratio	1.39	1.24	1.49	1.76

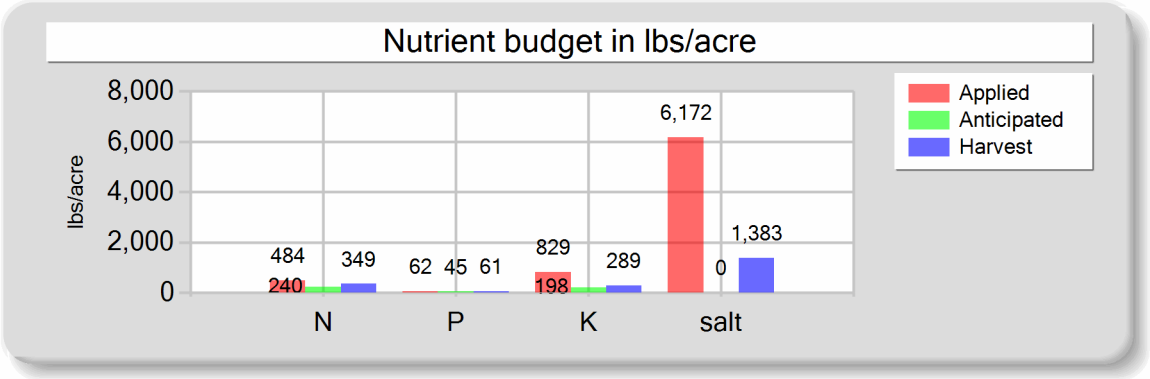
Fresh water applied	
15,561,490.00	*gallons*
573.08	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
1,935,000.00	*gallons*
71.26	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

A Silva - 05/25/2023: Corn, silage

Field name: A Silva Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	468.42	62.48	828.93	3,832.82
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	483.61	62.48	828.93	6,172.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	349.29	60.64	288.65	1,382.59
Nutrient balance	134.32	1.84	540.28	4,789.46
Applied to removed ratio	1.38	1.03	2.87	4.46

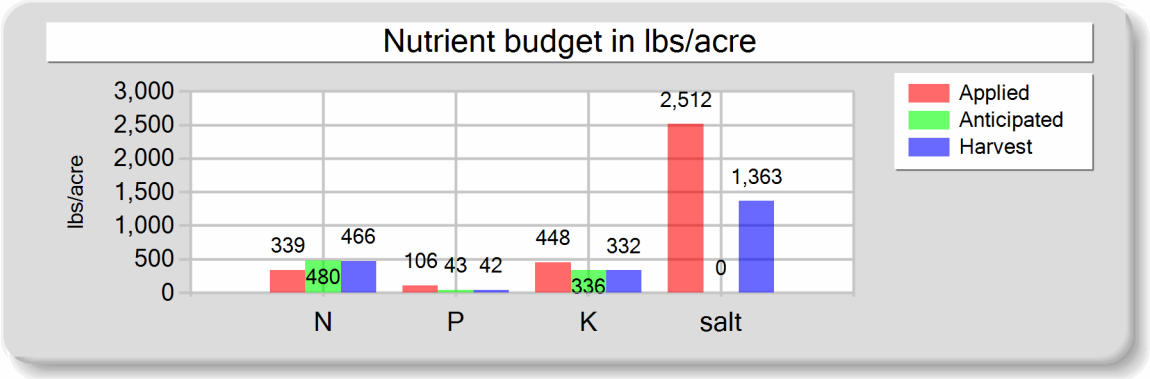
Fresh water applied	
85,496,252.00	*gallons*
3,148.54	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
4,905,000.00	*gallons*
180.63	*acre-inches*
2.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

G Silva - 11/04/2020: Alfalfa, hay

Field name: G Silva Crop: Alfalfa, hay Plant date: 11/04/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	316.65	105.55	447.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.79	0.00	0.00	2,511.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	339.44	105.55	447.78	2,511.62
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	466.44	41.62	331.53	1,363.44
Nutrient balance	-127.00	63.93	116.25	1,148.18
Applied to removed ratio	0.73	2.54	1.35	1.84

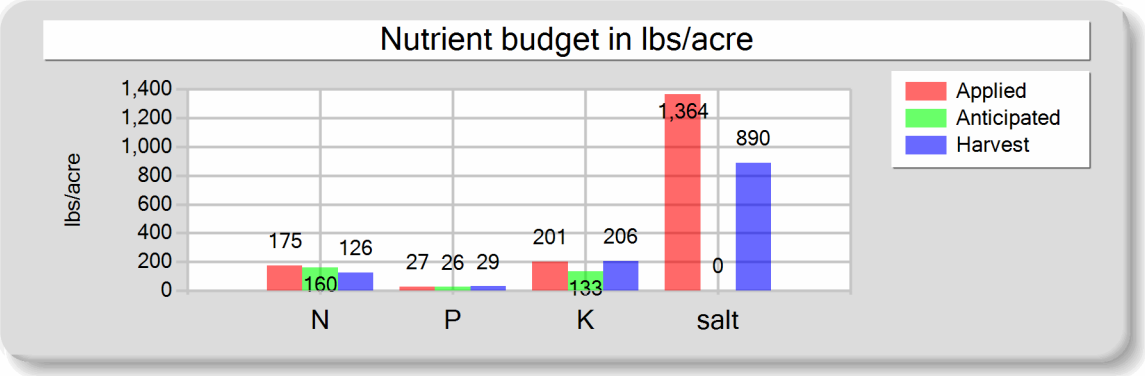
Fresh water applied	
58,689,739.00	*gallons*
2,161.34	*acre-inches*
55.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 10/16/2022: Oats, silage-soft dough

Field name: Home | Crop: Oats, silage-soft dough | Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.48	26.69	200.97	937.95
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.97	26.69	200.97	1,363.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.16	29.41	205.88	890.10
Nutrient balance	48.81	-2.72	-4.92	473.62
Applied to removed ratio	1.39	0.91	0.98	1.53

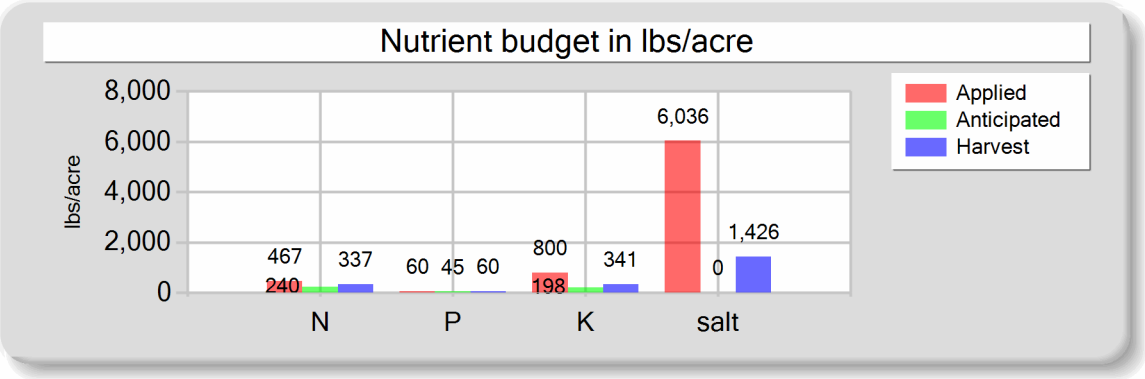
Fresh water applied	
22,194,257.00	*gallons*
817.34	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
2,182,500.00	*gallons*
80.37	*acre-inches*
0.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Home - 05/25/2023: Corn, silage

Field name: Home | Crop: Corn, silage | Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	452.08	60.33	799.65	3,697.24
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	467.27	60.33	799.65	6,036.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	336.96	59.53	341.45	1,426.46
Nutrient balance	130.31	0.80	458.20	4,610.01
Applied to removed ratio	1.39	1.01	2.34	4.23

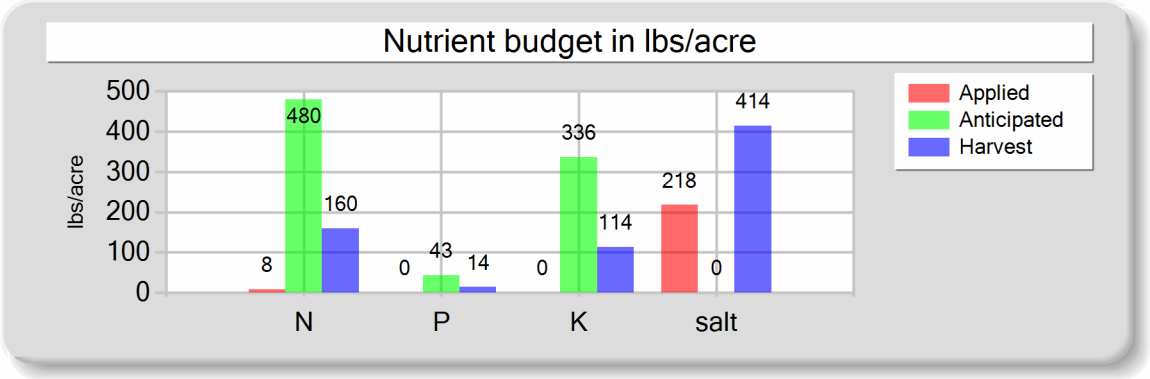
Fresh water applied	
121,937,279.00	*gallons*
4,490.54	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
6,750,000.00	*gallons*
248.58	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

M Silva - 11/04/2020: Alfalfa, hay

Field name: M Silva Crop: Alfalfa, hay Plant date: 11/04/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.76	0.00	0.00	217.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.76	0.00	0.00	217.76
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	160.12	13.96	113.57	414.25
Nutrient balance	-152.35	-13.96	-113.57	-196.50
Applied to removed ratio	0.05	0.00	0.00	0.53

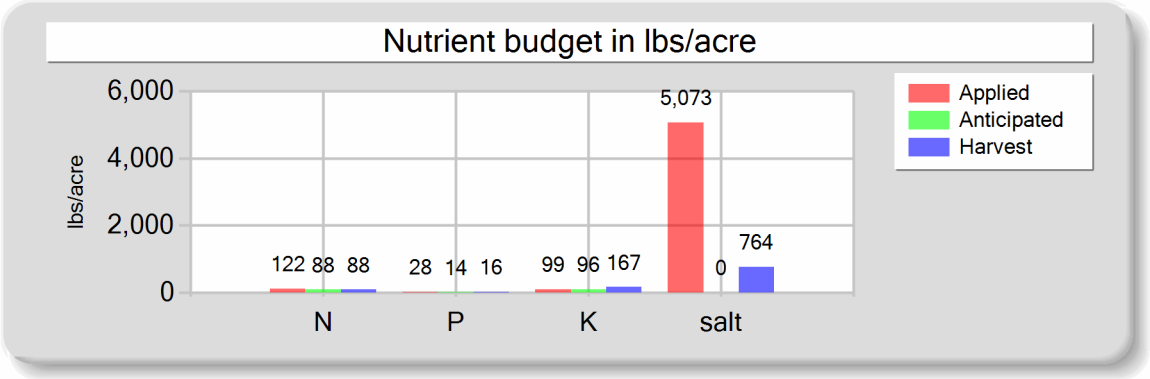
Fresh water applied	
10,046,255.00	*gallons*
369.97	*acre-inches*
4.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

M Silva - 05/25/2023: Sudangrass, silage

Field name: M Silva | Crop: Sudangrass, silage | Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.35	27.72	99.29	2,832.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.84	0.00	0.00	2,240.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	122.19	27.72	99.29	5,072.97
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	87.90	16.37	166.71	763.81
Nutrient balance	34.29	11.35	-67.42	4,309.16
Applied to removed ratio	1.39	1.69	0.60	6.64

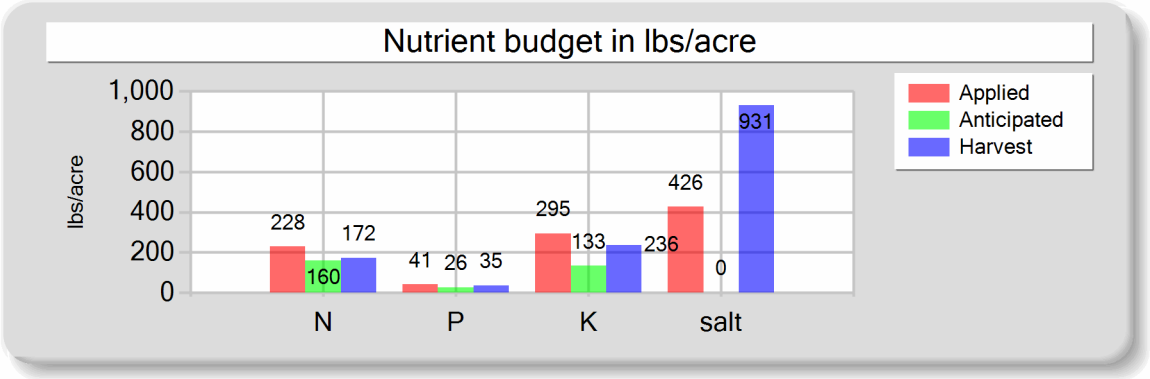
Fresh water applied	
103,366,122.00	*gallons*
3,806.62	*acre-inches*
49.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stephens - 10/16/2022: Oats, silage-soft dough

Field name: Stephens | Crop: Oats, silage-soft dough | Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	219.95	40.67	294.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.44	40.67	294.65	425.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	171.55	35.16	235.70	931.39
Nutrient balance	56.89	5.51	58.95	-505.62
Applied to removed ratio	1.33	1.16	1.25	0.46

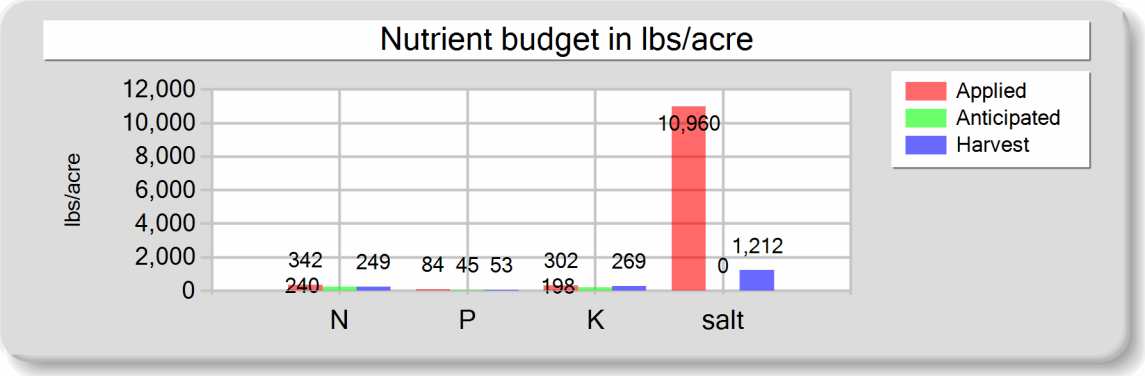
Fresh water applied	
5,867,447.00	*gallons*
216.08	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Stephens - 05/25/2023: Corn, silage

Field name: Stephens Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	326.72	84.37	302.18	8,620.59
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	341.91	84.37	302.18	10,959.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.58	53.41	269.13	1,212.10
Nutrient balance	93.33	30.95	33.05	9,747.73
Applied to removed ratio	1.38	1.58	1.12	9.04

Fresh water applied
32,236,291.00 *gallons*
1,187.15 *acre-inches*
51.62 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

SSB Manure - 2022

Sample and source description: SSB Manure - 2022

Sample date: 10/05/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,500.00	4,900.00	35,500.00							0.00
DL	100.00	100.00	100.00							1.00

SSB Manure

Sample and source description: SSB Manure

Sample date: 05/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,300.00	5,500.00	19,700.00	17,700.00	10,100.00	2,200.00	4,600.00	6,000.00		56.20
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

SSB Manure

Sample and source description: SSB Manure

Sample date: 10/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,700.00	9,900.00	42,000.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	557.00	434.00	0.00	5.40	72.70	886.00								9,190.00	4,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,140.00	427.00	0.00	1.50	209.00	1,070.00								10,400.00	4,600
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	742.00	453.00	0.00	0.00	107.00	1,170.00								11,000.00	5,320
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	687.00	464.00	0.00	1.20	90.00	1,250.00								10,700.00	5,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,260.00	469.00	0.00	0.00	232.00	1,170.00								12,300.00	4,650
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								0.01	1

Domestic Well

DW House

Sample description: DW House

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.20	0.00	14.20								1,330.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.80	0.00	5.80								1,160.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

A Silva - 10/16/2022: Oats, silage-soft dough

Oats A Silva

Sample and source description: Oats A Silva

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,600.00	23,000.00		8.10
DL	0.01	0.01	0.01		1.00

A Silva - 05/25/2023: Corn, silage

A Silva - 05/25/2023: Corn, silage

A Silva Corn

Sample and source description: A Silva Corn

Sample date: 10/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,500.00	11,900.00		5.70
DL	0.01	0.01	0.01		1.00

G Silva - 11/04/2020: Alfalfa, hay

G. Silva Alfalfa

Sample and source description: G. Silva Alfalfa

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,500.00	2,900.00	23,100.00		9.50
DL	0.01	0.01	0.01		1.00

Home - 10/16/2022: Oats, silage-soft dough

Home B, C, D Oats

Sample and source description: Home B, C, D Oats

Sample date: 04/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,800.00	26,600.00		11.50
DL	0.01	1.00	0.01		1.00

Home - 05/25/2023: Corn, silage

Corn Home A

Sample and source description: Corn Home A

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	2,800.00	18,100.00		7.20
DL	0.01	0.01	0.01		1.00

Corn Home B/C

Sample and source description: Corn Home B/C

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,500.00	12,300.00		5.50
DL	0.01	0.01	0.01		1.00

M Silva - 11/04/2020: Alfalfa, hay

M. Silva Alfalfa

Sample and source description: M. Silva Alfalfa

Sample date: 05/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,400.00	3,000.00	24,400.00		8.90
DL	0.01	0.01	0.01		1.00

M Silva - 05/25/2023: Sudangrass, silage

M Silva - 05/25/2023: Sudangrass, silage

M Silva-Sudan Silage

Sample and source description: M Silva-Sudan Silage

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,700.00	27,500.00		12.60
DL	0.01	0.01	0.01		1.00

Stephens - 10/16/2022: Oats, silage-soft dough

Stephens Oat Silage

Sample and source description: Stephens Oat Silage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,050.00	3,700.00	24,800.00		9.80
DL	100.00	100.00	100.00		1.00

Stephens - 05/25/2023: Corn, silage

Stephens Corn Silage

Sample and source description: Stephens Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,600.00	13,100.00		5.90
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

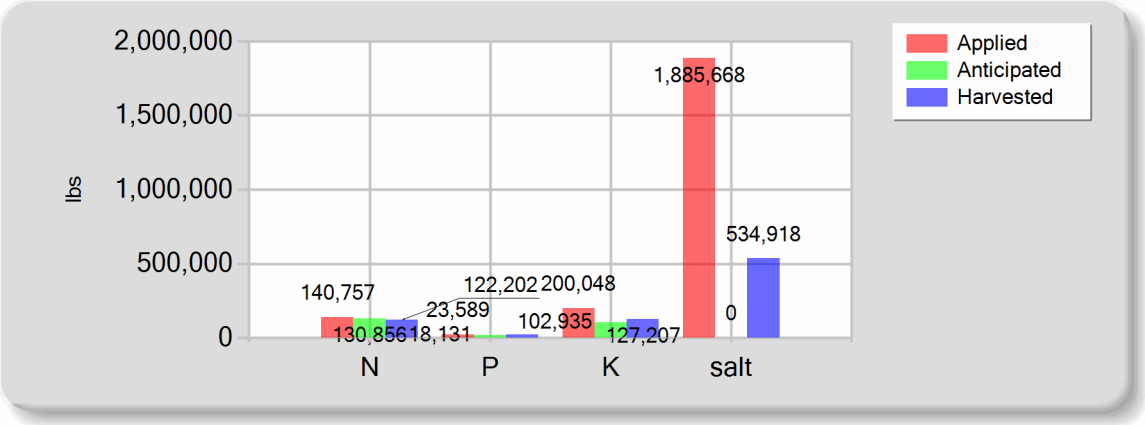
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

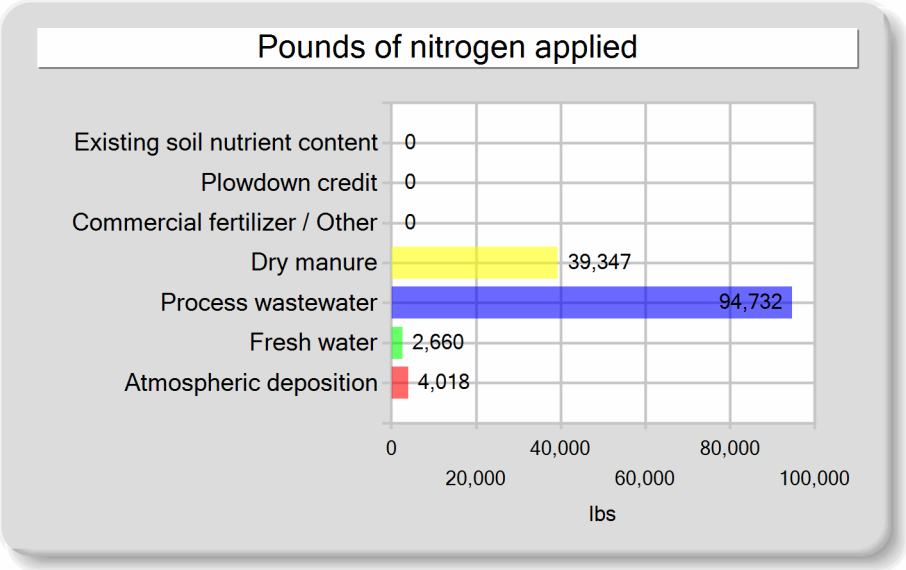
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

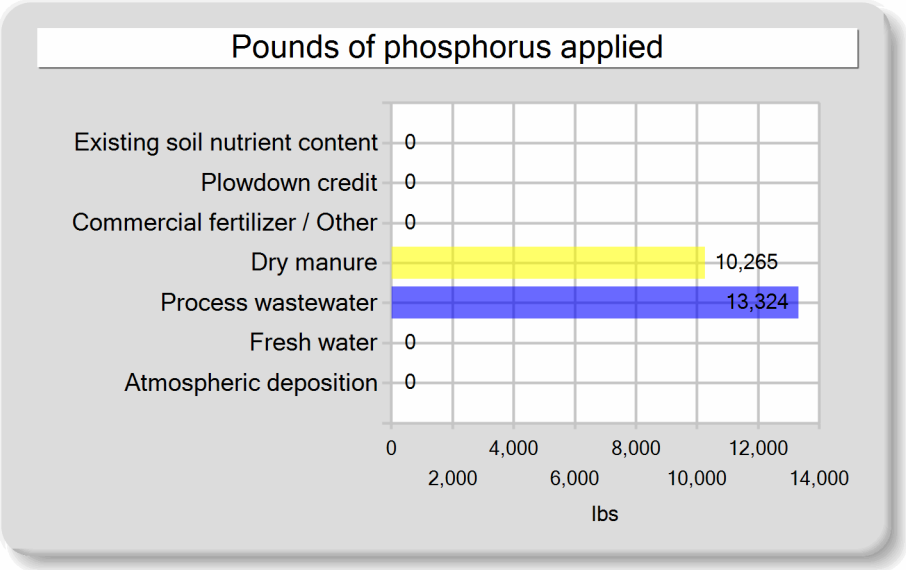
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39,347.45	10,265.43	47,086.09	416,374.56
Process wastewater	94,731.72	13,323.78	152,961.47	709,239.05
Fresh water	2,660.19	0.00	0.00	760,054.48
Atmospheric deposition	4,018.00	0.00	0.00	0.00
Total nutrients applied	140,757.36	23,589.21	200,047.56	1,885,668.08
Anticipated crop nutrient removal	130,856.00	18,131.00	102,934.80	0.00
Actual crop nutrient removal	122,202.38	19,915.34	127,207.19	534,917.75
Nutrient balance	18,554.99	3,673.87	72,840.36	1,350,750.33
Applied to removed ratio	1.15	1.18	1.57	3.53

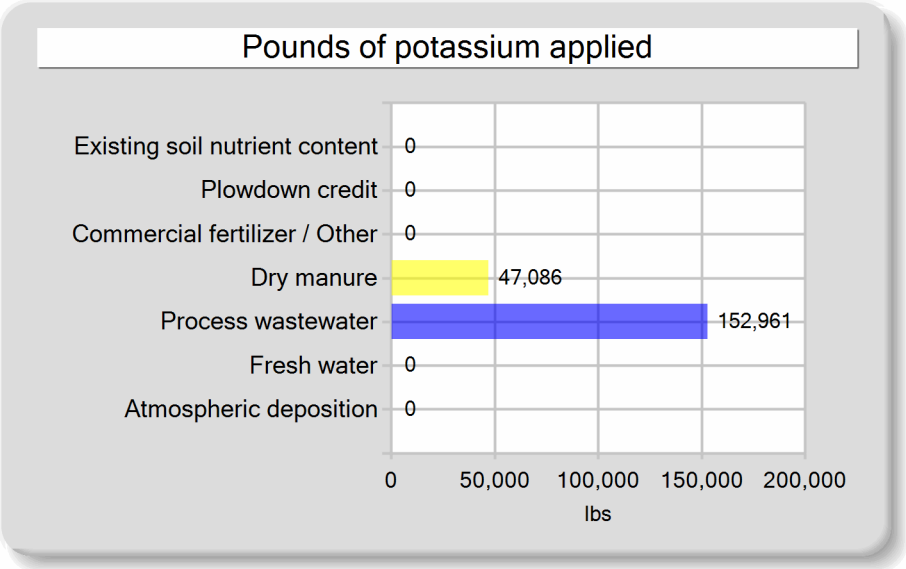
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

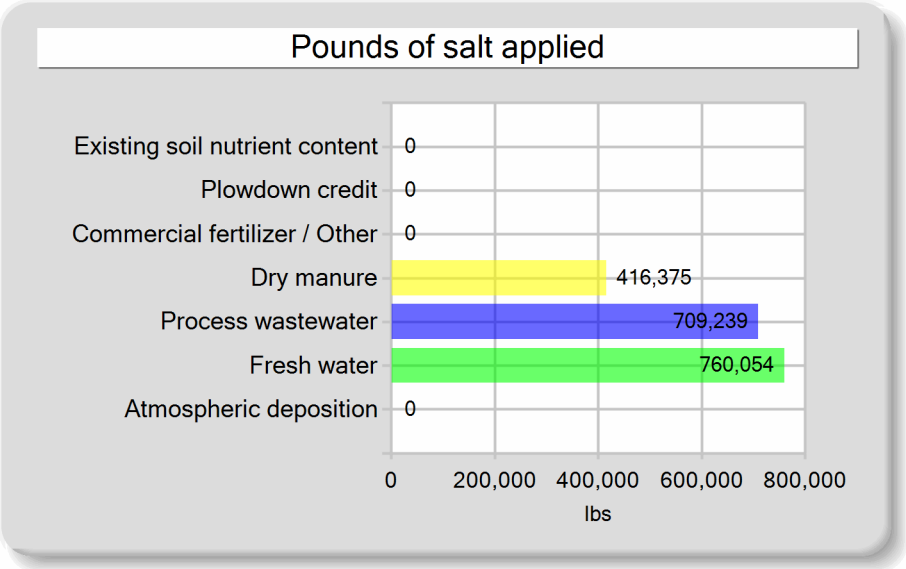


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Luis Nunes	Louie Nunes
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Luis C Nunes	Louie Nunes
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 28, 2024	Jun 28, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Louie Nunes

Name of Dairy Facility: Luis C Nunes & Sons Dairy #1

Facility Address:

29191 Husman RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Louie Nunes (209) 652-7647
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Luis C Nunes Dairy #1

Address of Hauling Company/Person:

29191 Husman RD	Gustine	CA	95322
Number and Street	City	State	Zip Code

Contact Person: Louie Nunes (209) 652-7647
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Luis C Nunes Dairy #2 (209) 652-7647
Name Phone Number

10268 Hwy 33	Gustine	CA	95322
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

	Gustine	95322
Address	City	Zip Code

Hwy 33	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 04/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,575.00 tons

Manure Solids Content: 50.4 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Louie Nunes 4/18/23

Operator Signature Date

Louie Nunes 4/18/23

Hauler Signature Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Louie Nunes

Name of Dairy Facility: Luis C Nunes & Sons Dairy #1

Facility Address:

29191 Husman RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Louie Nunes (209) 652-7647
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Luis C Nunes Dairy #1

Address of Hauling Company/Person:

29191 Husman RD	Gustine	CA	95322
Number and Street	City	State	Zip Code

Contact Person: Louie Nunes (209) 652-7647
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Luis C Nunes Dairy #2 (209) 652-7647
Name Phone Number

10268 Hwy 33	Gustine	CA	95322
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

	Gustine	95322
Address	City	Zip Code

Hwy 33	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 10/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,300.00 tons

Manure Solids Content: 27.0 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Louie Nunes 10/18/23

Operator Signature Date

Louie Nunes 10/18/23

Hauler Signature Date

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Luis C Nunes Dairy #1
29191 Husman Road
Gustine CA 95322
00-0021362 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0994
Sampled By: M DeKeyrel
Requested By: Louie Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	12:05	1160	1.16	5.8	ND		7.9
2	DW - House	12/13/23	12:11	1330	1.33	14.2	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Luis C Nunes Dairy #1
29191 Husman Road
Gustine CA 95322
00-0021362 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0994
Sampled By: M DeKeyrel
Requested By: Louie Nunes
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	12:05	**1160**	**1.16**	**5.8**	15.83	1.32
2	DW - House	12/13/23	12:11	**1330**	**1.33**	**14.2**	38.77	3.23

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0994-01	DW - MB	Ag Water	M DeKeyrel		12/13/2023 12:05
23L0994-02	DW - House	Ag Water	M DeKeyrel		12/13/2023 12:11

Default Cooler Temperature on Receipt °C: 4.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Sample Results

Sample: DW - MB
23L0994-01 (Water)

Sampled: 12/13/2023 12:05
Sampled By: M DeKeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.16**	mmhos/cm	0.01	1		12/18/23 17:50	SM 2510 B		BEL0704
Electrical Conductivity umhos	**1160**	umhos/cm	10.0	1		12/18/23 17:50	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:05	Field		BEL0633
Nitrate Nitrogen as NO3N	**5.8**	mg/L	0.1	1	10	12/17/23 07:58	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:50	SM 4500-H+	H	BEL0704
pH	**7.9**	units	1.0	1		12/18/23 17:50	SM 4500-H+	H	BEL0704

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Sample Results
(Continued)

Sample: DW - House
23L0994-02 (Water)

Sampled: 12/13/2023 12:11
Sampled By: M DeKeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.33**	mmhos/cm	0.01	1		12/18/23 17:51	SM 2510 B		BEL0704
Electrical Conductivity umhos	**1330**	umhos/cm	10.0	1		12/18/23 17:51	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:11	Field		BEL0633
Nitrate Nitrogen as NO3N	**14.2**	mg/L	0.1	1	10	12/17/23 08:18	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:51	SM 4500-H+	H	BEL0704
pH	**7.6**	units	1.0	1		12/18/23 17:51	SM 4500-H+	H	BEL0704

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DELLAVALLE™ LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665										
Blank (BEL0665-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0665-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.2	90-110		
LCS (BEL0665-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
LCS (BEL0665-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.0	90-110		
LCS (BEL0665-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEL0665-DUP1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.639	10
Duplicate (BEL0665-DUP2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			0.787	10
Duplicate (BEL0665-DUP3)		**Source: 23L1000-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.612	10
Duplicate (BEL0665-DUP4)		**Source: 23L1005-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.7			0.941	10
Matrix Spike (BEL0665-MS1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L	5.000	5.8	102	90-110		
Matrix Spike (BEL0665-MS2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.3	102	90-110		

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DELLAVALLE™ LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665 (Continued)										
Matrix Spike (BEL0665-MS3)			**Source: 23L1000-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.2	91.6	90-110		
Matrix Spike (BEL0665-MS4)			**Source: 23L1005-02**		Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.7	99.3	90-110		
Reference (BEL0665-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0665-SRM2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0665-SRM5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704										
Blank (BEL0704-BLK1)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0704-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0704-BLK3)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Duplicate (BEL0704-DUP1)		**Source: 23L0999-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.01		0.01	mmhos/cm		1.01			0.248	10
pH	7.7		1.0	units		7.6			0.784	10
Electrical Conductivity umhos	1010		10.0	umhos/cm		1010			0.248	10
Duplicate (BEL0704-DUP2)		**Source: 23L1004-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.11		0.01	mmhos/cm		1.11			0.108	10
Electrical Conductivity umhos	1110		10.0	umhos/cm		1110			0.108	10
pH	7.8		1.0	units		7.7			0.775	10
Reference (BEL0704-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	426			umhos/cm	426.0		99.9	90-110		
Reference (BEL0704-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0704-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #1
29191 Husman Road
Gustine, CA 95322

Account# 00-0021362
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:13

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704 (Continued)										
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0704-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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12/14/23 14:20 23L0994

MMM

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 ~~800 228-9896~~ • Fax 559 268-8174

WATER WORK REQUEST

Acct No. Cons.
Bill To: 21362 8

Purchase Order No. Results Needed By

Client: Nunes Dairy #1, Luis C
Address: 29191 Husman Road
City, State, Zip: Gustine, CA 95322
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Louie Nunes - (209) 652-7647

Ranch:

Date sampled: 12/13/23

Sampled by: M Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: No. Bottles:

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[X] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

[X] DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	12:05	Ø	4.9
2. DW-House	Sampled From:	12/13	12:11	Ø	4.2
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M. Dekeyrel	CAS Inc	12/13/23 @ 12:11	12/14/23 12p
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___ Date ___

Shipping
$ ___ In
$ ___ Out

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/14/23 14:20 23L0994

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT